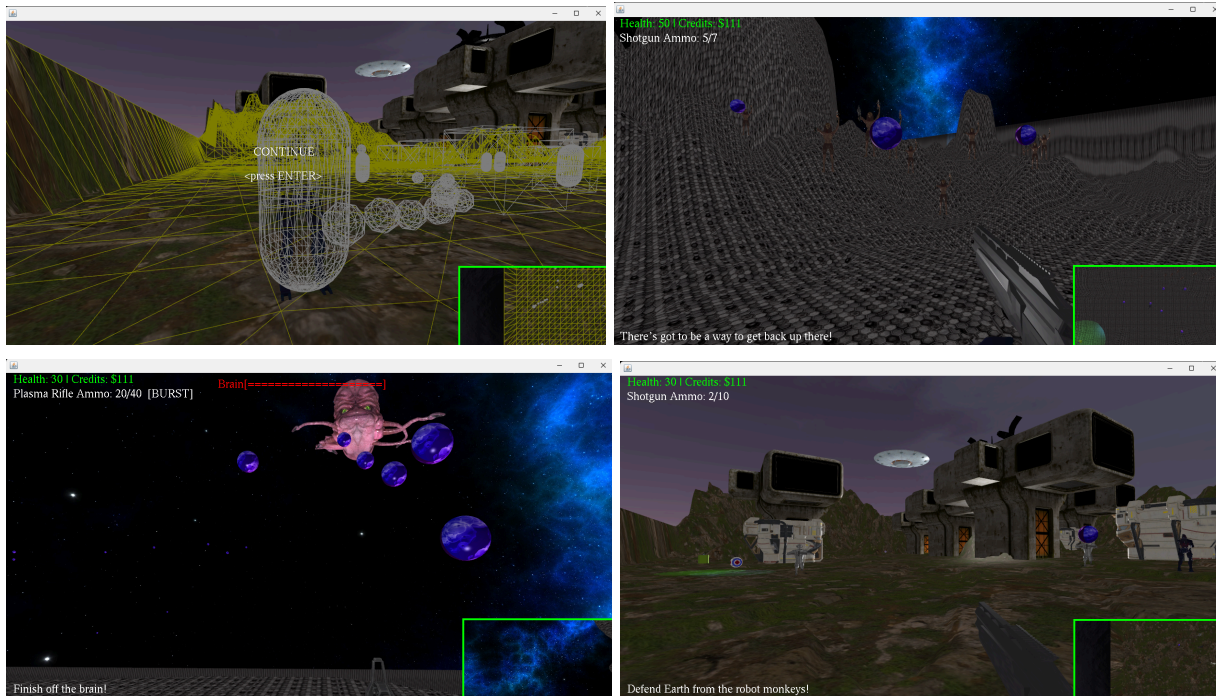


1. Robo-Rock by Ryan Lee and Nick Pasciuti CSC165-01 Spring 2026
2. Screenshot of our game:



3. To compile our game, in order to run:
 - buildTAGE.bat
 - compile.bat
 - server.bat | *not necessary to play even in multiplayer.*
 - run.bat
4. Aliens are invading, and you need to fend off waves of robot apes. Once you clear a group of apes, another UFO will drop more apes until the mothership appears. Finish off the last apes and ride the tractor beam to take the fight to their planet. A giant brain knocks you down to the alien-infested field below. Pick them off until one drops the grapple, use it to get back up to the brain, and start the final battle. Defeat the brain, and you win!
5. Controls:
 - Up Arrow / Gamepad Left Stick Up: Move menu selection up or Move Vending selection up
 - Down Arrow / Gamepad Left Stick Down: Move menu selection down or Move Vending selection down
 - Left Arrow / Gamepad Left Stick Left : Move menu selection left
 - Right Arrow / Gamepad Left Stick Right : Move menu selection right
 - Enter / Gamepad A Button: Confirm menu option / buy selected vending item
 - A or Left Arrow / Gamepad Left Stick Left: Choose previous robot on robot selection screen

- D or Right Arrow / Gamepad Left Stick Right: Choose next robot on robot selection screen
 - Enter / Gamepad A Button: Confirm robot and start game
 - Escape: Return to the main menu from the robot selection screen
 - W / Gamepad Left Stick Up: Move forward
 - S / Gamepad Left Stick Down: Move backward
 - A / Gamepad Left Stick Left: Strafe left
 - D / Gamepad Left Stick Right: Strafe right
 - Mouse Movement / Gamepad Right Stick: Look and aim
 - Left Mouse Button / Gamepad Right Trigger or Right Bumper: Fire weapon
 - R / Gamepad X Button: Reload
 - Keyboard E / Gamepad B Button: Grapple / Enter Vending Machine / Exit Vending Machine
 - Mouse Wheel / Gamepad Y Button: Switch to next weapon
 - Mouse Wheel / Gamepad Left Bumper: Switch to previous weapon
 - Keyboard B: Toggle plasma rifle fire mode
 - F: Toggle first-person mode
 - Escape: Pause game
 - Numpad +: Zoom overhead camera in
 - Numpad -: Zoom overhead camera out
 - 1: Vending full health | Buy with A on gamepad
 - 2: Vending full ammo | Buy with A on gamepad
 - 3: Buy damage multiplier upgrade (up to 4 times) | Buy with A on gamepad
 - 4: Buy ammo multiplier upgrade (up to 4 times) | Buy with A on gamepad
 - Arrow Keys: Pan the overhead camera
 - C: Recenter overhead camera
 - P: Toggle physics debug view | DEBUG Option
 - Enter / Gamepad A Button: Continue after death (up to 3 lives)
 - Backslash: Restart game / return to menu
 - T: Queue final UFO tractor beam test | DEBUG Option
 - Y: Spawn debug skinny loadout | DEBUG Option
 - U: Force level two arrival event | DEBUG Option
6. We have a spotlight below the ammo and one below health pickups, both of which turn off when ammo or health is picked up and back on when the ammo or health respawns after 30 seconds. We also have a spotlight that acts as the UFO's tractor beam.
 7. We used a UDP client-server based on TAGE's networking. The create and move messages now include our avatar. We send the player's ID, position, and selected avatar. Remote clients use the avatar value to assign the correct texture to ghosts.
 8. We changed the HUD manager to include a fourth HUD.
 9. Genre, theme, dimensionality, and activities:
 - Genre: First-person shooter (FPS)
 - Theme: Sci-fi alien invasion with robot defenders
 - Dimensionality: 3D game world with terrain, skyboxes, physics-based movement, animated models, HUD, and multiple levels.

- Activities: The player fights waves of enemy NPCs, collects ammo and health, earns credits from defeated enemies, uses multiple weapons, enters a tractor beam, transitions to a second map, gets thrown off a cliff, uses a grapple, and fights a boss.

10. Where each project requirement is visible in the game:

- External models are visible throughout the game as weapons, buildings, pickups, enemies, UFOs, and grapple.
- Networked multiplayer is visible when a client creates a joinable game by selecting Multiplayer, then Host Game from the start menu. A second player can select Join Game, and see each other's ghost avatars moving in the world, UFO and NPC ghost positions update to the first client, and attach to whoever is closer. A terminal window will display diagnostic data about the two or more connected clients.
 - A separate server not hosted by the client can also be started using server.bat, then selecting the join option on any clients running the game.
- Avatar selection is visible on the robot selection screen before the game starts. The selected robot texture is sent over the network and displayed on remote ghost avatars.
- Skybox and terrain are visible in both playable areas. The first area uses terrain and skybox scenery, and the second area switches to a different terrain and skybox setup.
- Lights are visible through the main world lighting, pickup spotlights, and the UFO tractor beam light. We satisfied the requirement of having lights turn on and off by having the spotlight below the pickups turn off when an item was picked up, and back on when an item respawned.
- The HUD is visible during menus, robot selection, gameplay, death/continue screens, weapon display, health, credits, objectives, and boss health.
- 3D sound is used for action sounds, such as pickups, enemy weapon fire, and enemy death.
- Hierarchical scene graph usage is evident in weapons attached to the player and enemy characters. The weapons follow their parent character using TAGE's scene graph propagation.
- Animation is visible on the player robot, ghost avatars, apes, skinnys, and the brain boss.
- NPC/AI behavior is visible through enemy apes, skinnys, and the brain boss. The ape enemies use behavior tree actions such as facing the player, moving based on distance, strafing, and firing.
- Physics is visible through player movement, terrain collision, bullets, enemy physics, building colliders, gravity, and physics debug mode.
- Keyboard and gamepad input are both supported for movement, aiming, firing, menu navigation, robot selection, and all gameplay actions. Debug keys are not mapped to the gamepad.
- Single-player works and lets the player explore the world, fight enemies, collect pickups, and experience the main game features without another player.

11. We were able to get all the requirements working.
12. Some extra things include multiple weapon types, reload and ammo systems, health and ammo pickups with respawn timers, UFO waves, a tractor beam transition, a second playable map, a grapple pickup and grapple movement ability, boss music, a brain boss encounter, a continue screen, and credits earned from defeated enemies.
13. Contributions from each team member:
Ryan Lee designed and implemented the weapon scene graph, NPCs, and networking. Created the ammo and health model and texture. Worked on modeling, behavior trees, rigging animations, avatar selection, 3d sounds.
Nick Pasciuti designed terrain heightmaps, score systems, vending, menus, and networking. Created the knife model and texture. Worked on modeling, scoring, menu selection, bullets, behavior trees, gun behavior, and multiplayer menu.
14. Assets created by Ryan Lee:
 - ammo.obj, ammo.jpg
 - health.obj, health.jpg
 - pistol.jpgAssets created by Nick Pasciuti:
 - knife.obj, knife.png
 - Heightmaps map0hm.png and map1hm.png
15. Outside Sources and Licenses:

Textures:

Map 0 Terrain | coast_sand_rocks_02_diff_1k.jpg

[Coast Sand Rocks 02 Texture • Poly Haven](#) - CC0

[License • Poly Haven](#)

Map 1 Terrain | airbase_radar_panels.jpg

[Golgotha Textures: airbase radar panels.jpg - airbase radar panels.jpg | OpenGameArt.org](#)

[Deed - CC0 1.0 Universal - Creative Commons](#)

Bullets: https://www.freepik.com/free-vector/gold-metal-background_34294136.htm credit to juicy_fish

Plasma Bullets:

https://www.freepik.com/free-vector/abstract-hand-painted-alcohol-ink-background-design_50500296.htm credit to kjpargeter

Crosshairs: <https://www.hiclipart.com/free-transparent-background-png-clipart-mryvr>

License

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Brain: Texture was saved from Blender after importing LostBoyz2078 .glb file for Brain Run and finding its base color in the shading tab. brain.jpg. See links below for more info.

Futuristic Skyscraper was used in the same process, credit to MOjackal to make centerBuilding.jpg

Scifi Buildings Pack used the same process credit to Golukumar.

Scifi Building texture was derived the same way as well.

<https://pinetools.com/invert-image-colors> image inverter was used to make robot1,2,3.jpg

Models:

robocop used to create robot.obj and robot.jpg -

<https://sketchfab.com/3d-models/robocop-b37bee32032d468c9c1cbd041e8142c1>

plasma-rifle used to create plasmaRifle.obj and plasmaRifle.jpg -

<https://sketchfab.com/3d-models/plasma-rifle-bf439e8a1f8e41888d7e94592c352807>

Pistol | pistol.obj

<https://www.turbosquid.com/FullPreview/922989>

Terminal/VendingMachine - Credit to Tronin Dmitry | Terminal UV_1.obj

uv_checker_material_uv_grid_2048x2048_Roug.png

[Sci-fi Terminal - Download Free 3D model by Tronin Dmitry \(@kosmotron\) \[78ac943\]](#)

Skinwalker_creature skinny.obj, skinny.jpg

<https://sketchfab.com/3d-models/skinwalker-creature-1536d87ab17b4d9e889d9195f5ceca0b>

Apeman Futura ape.obj, ape.jpg

<https://sketchfab.com/3d-models/apeman-futurastargate-5c334a1f9d8e4f87b62be7e62e71606e>

low -poly Vertical DBS was used for shotGun.obj and shotGun.jpg

<https://sketchfab.com/3d-models/low-poly-vertical-dbs-709ba08245a24b34b1495e4f46255e9f>

Credit to Streljay_Ataman

Futuristic Modular Rifle was used for rifle.obj and rifle.jpg

<https://sketchfab.com/3d-models/futuristic-modular-rifle-2389f96cfac442b58272a489d7cdd606>

Credit to Ramhat

UFO <https://sketchfab.com/3d-models/ufo-75655f43c4fc4c56ab60746caf7119fe> Credit to

slebsom ufo.obj, ufo.png

Futuristic Skyscraper: centerBuilding.obj centerBuilding.jpg

<https://sketchfab.com/3d-models/futuristic-skyscraper-b79ec00a53cf4b72ac0e8d6f055bc21c>

Credit to MOjackal

Sci-Fi-Buildings Pack smallBuilding.obj smallBuilding.png

<https://sketchfab.com/3d-models/sci-fi-buildings-pack-cc74d81f4e0e4db3acf6f3433c678250>

Credit to Golukumar

SciFi Building smallBuilding2.obj smallBuilding2.png

<https://sketchfab.com/3d-models/sci-fi-building-0cc82a32dfb84e858f76813e1c42b268>

Grapple: <https://sketchfab.com/3d-models/grapple-gun-aeac4217c98a4bb489688a65fe831d74>

Credit to AI Garcia grapple.obj, grapple.png

Brain: <https://sketchfab.com/3d-models/brain-run-a84217d7f44146d2ad22c018ab9976ed>

License CC non-commercial credit to LostBoyz2078 brain.obj, brain.jpg

Animations were made from robot.obj, ape.obj, skinny.obj, and brain.obj using Professor Gordon's .rkm, .rks, .rka file exporter.

Sounds:

Ammo Pick-Up <https://freesound.org/people/Dpoggioli/sounds/213607/> ammoPickup.wav

Pick-Up Health <https://freesound.org/people/juancamiloorjuela/sounds/204318/>

healthPickup.wav

Rifle <https://freesound.org/people/bolkmar/sounds/455922/> in a loop by bolkmarm. rifle.wav

WeaponSwap: <https://freesound.org/people/twisterman/sounds/159450/> Credit to twisterOrtiz
Plasma Rifle is Robo gun:

<https://freesound.org/people/newlocknew/sounds/514074/> Credit to Newlocknew

plasmaRifle.wav

Shotgun, Pistol: <https://f8studios.itch.io/snakes-authentic-gun-sounds> by SnakeF8 pistol.wav,
shotGun.wav, sgPump.wav

Apes Laser Gun: <https://pixabay.com/sound-effects/film-special-effects-laser-zap-90575/>

<https://pixabay.com/service/license-summary/> npcPlasma.wav

Apes Death: <https://pixabay.com/sound-effects/film-special-effects-death-408455/> apeDie.wav

Alien Death: <https://pixabay.com/sound-effects/film-special-effects-alien-high-pitch-312010/>

alienDie.wav

Brain Roar: <https://pixabay.com/sound-effects/film-special-effects-epic-dragon-roar-364481/>

roar.wav

Die: <https://pixabay.com/sound-effects/film-special-effects-retro-video-game-death-95730/>

die.wav

Game Music:

Main Menu/Level 1

<https://pixabay.com/music/video-games-90s-game-music-no-copyright-352850/> level1.wav

Level 2

<https://pixabay.com/music/video-games-spaceship-arcade-shooter-game-background-soundtrack-318508/> level2.wav

Boss

<https://pixabay.com/music/upbeat-video-game-boss-fight-259885/> boss.wav

All pixabay game music is free to use.

Skyboxes: <https://screamingbrainstudios.itch.io/planet-surface-skyboxes> License - CC0

[Space Skyboxes | OpenGameArt.org](https://opengameart.org/) License - CC0

[Deed - CC0 1.0 Universal - Creative Commons](https://creativecommons.org/licenses/by/4.0/)

We used <https://screamingbrainstudios.itch.io/cubemap-splitter> tool to create the 6 images

This utility has been released under the Public Domain (CC0) license and can be distributed freely without restrictions.

16. This game was tested and confirmed to run on:

- ECS - RAYMAN
- ECS - SNEEZYMUD